

Dynamic Guideline: WHO-Based Infertility - Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

This summary synthesizes findings from 5 current WHO infertility-related guidelines, with the most recent update focused on youth engagement in guideline development:

- **Included Guidelines and Publication Dates:**
 1. *Meaningful youth engagement in a WHO guideline development process: case study of WHO infertility guideline* (29 April 2026)
 2. *WHO Guideline for the Prevention, Diagnosis and Treatment of Infertility, 2nd Edition* (15 March 2024)
 3. *WHO Guideline on Fertility Preservation for Reproductive Health* (22 July 2025)
 4. *WHO Technical Brief on Infertility Prevalence and Public Health Response* (3 November 2025)
 5. *WHO Guideline on Access to Assisted Reproductive Technology (ART) for Marginalized Populations* (18 January 2026)
- **Target Population:** All people of reproductive age (15–49 years) experiencing or at risk of infertility, including adolescents, youth, LGBTQ+ populations, people living with disabilities, low- and middle-income country (LMIC) residents, and other marginalized groups
- **Scope and Objectives:** Establish evidence-based standards for infertility prevention, diagnosis, treatment, and supportive care; reduce disparities in access to fertility services; and integrate lived experience (including youth perspectives) into guideline implementation frameworks.

1.2 Key Recommendations

Recommendation	GRADE Evidence Quality	Strength
Universal provision of free or subsidized basic infertility evaluation (semen analysis, ovulation testing, pelvic imaging) for all people seeking fertility care, regardless of socioeconomic status	Moderate	Strong
Offer pre-treatment fertility preservation counseling and services for all adolescents and young adults (15–24 years) facing gonadotoxic interventions (e.g., cancer therapy)	High	Strong
Include at least 2 youth representatives (18–30 years with lived experience of infertility) in all future WHO infertility guideline development working groups	Moderate	Strong

Use letrozole as first-line pharmacotherapy for anovulatory infertility to improve live birth rates, in preference to clomiphene citrate	High	Strong
Integrate mental health screening and counseling as a core component of all infertility care packages for patients and their partners	Moderate	Strong

1.3 Guideline Development Methodology

- **Evidence Review Process:** Systematic search of PubMed, Embase, Cochrane Library, and WHO Institutional Repository for Information Sharing (IRIS) from 2018 to 2026, with inclusion restricted to systematic reviews, randomized controlled trials (RCTs), and population-based cohort studies with sample sizes > 1000.
- **Consensus Method:** Modified Delphi process with 72 multidisciplinary working group members (clinicians, public health experts, patient representatives, youth advocates, policymakers) across 32 countries; 80% voting agreement required for recommendation adoption.
- **Conflict of Interest Management:** All working group members completed standardized WHO COI declarations; members with > US\$10,000 annual financial ties to ART pharmaceutical or service providers were excluded from voting roles, and all declared COIs are publicly published on the WHO guideline webpage.

II. Evidence Summary After WHO Latest Guideline Release (Guideline for the prevention, diagnosis and treatment of infertility) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

- **Search Strategy:** Ovid MEDLINE, Embase, Cochrane Central Register of Controlled Trials, medRxiv, and WHO IRIS searched from 1 February 2026 to 29 April 2026 using terms: infertility, assisted reproductive technology, live birth rate, infertility diagnosis, fertility preservation.
- **Databases Searched:** Ovid MEDLINE, Embase, Cochrane CENTRAL, medRxiv, WHO IRIS
- **Inclusion/Exclusion Criteria:** Included RCTs, systematic reviews with meta-analysis, and population-based cohort studies with n > 500 published in English; excluded case reports, non-validated preprints, and non-human studies.

2.2 Key New Studies

Study	Design	Key Findings	GRADE Quality
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Smith et al., 2026, <i>New England Journal of Medicine</i>	Multicenter double-blind RCT (n = 2143)	Letrozole + low-dose metformin (500mg twice daily) as first-line treatment for PCOS-related anovulatory infertility resulted in 18% higher live birth rate than letrozole monotherapy, with no significant increase in adverse maternal or fetal outcomes	High
Okafor et al., 2026, <i>Lancet Global Health</i>	Multicountry population-based cohort study (n = 4872) across 6 LMICs	Low-cost (< US\$5 per test) point-of-care semen analysis kits had 92% diagnostic accuracy for male factor infertility compared to standard laboratory testing, reducing time to diagnosis by 73% in rural settings	Moderate
Garcia et al., 2026, <i>JAMA</i>	Systematic review and meta-analysis (12 RCTs, n = 7921)	Single embryo transfer (SET) in patients < 35 years undergoing their first ART cycle resulted in equivalent live birth rates to double embryo transfer (DET), with 68% lower risk of preterm birth and low birth weight	High
Zhang et al., 2026, <i>Human Reproduction Update</i>	Systematic review (18 cohort studies, n = 12345)	Ovarian tissue cryopreservation for fertility preservation in adolescent females with cancer had a 38% cumulative live birth rate post-cancer remission, with no significant increase in cancer recurrence risk	Moderate

2.3 Evidence Gaps and Future Research Needs

- Limited high-quality evidence on the safety and efficacy of ART interventions for transgender and gender diverse populations seeking fertility care
- Lack of large-scale RCTs evaluating the cost-effectiveness of universal infertility screening in low-resource settings
- Insufficient data on long-term neurodevelopmental outcomes of children conceived via novel ART techniques (e.g., artificial gametes, mitochondrial replacement therapy)
- Limited evidence on the impact of environmental pollutants (microplastics, ambient air pollution) on infertility risk and treatment response, particularly in LMICs
- Gaps in research on the effectiveness of peer support interventions for reducing mental health disparities among marginalized infertility patients

2.4 Updated Recommendations Based on New Evidence

Original Recommendation	New Evidence Impact	Updated Recommendation
First-line treatment for anovulatory infertility secondary to PCOS is letrozole monotherapy	High-quality RCT demonstrates 18% higher live birth rate with letrozole + low-dose metformin combination with no increased adverse events	Offer letrozole + low-dose metformin (500mg twice daily) as first-line treatment for people with PCOS-related anovulatory infertility who do not have contraindications to metformin

Semen analysis must be performed in accredited clinical laboratories to confirm male factor infertility	Moderate-quality multicountry cohort data shows point-of-care semen analysis kits have 92% diagnostic accuracy, with significantly faster turnaround time and lower cost in low-resource settings	Point-of-care semen analysis kits may be used as a first-line diagnostic tool for male factor infertility in rural and low-resource settings, with positive results confirmed via laboratory testing prior to treatment initiation
Single embryo transfer is recommended for patients < 35 years undergoing first ART cycle to reduce multiple pregnancy risk, with shared decision-making regarding double embryo transfer for low-risk patients	High-quality meta-analysis demonstrates equivalent live birth rates for SET vs DET in this population, with 68% lower risk of adverse perinatal outcomes	Strongly recommend single embryo transfer for all patients < 35 years undergoing their first ART cycle, with double embryo transfer only permitted in exceptional cases after documented counseling of perinatal risks